# AZHAD SHAHZAD SHAIK


Engineer & Applied Researcher | Security, Machine Learning & Intelligent Systems

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## RESEARCH INTERESTS

- Critical-infrastructure and EV/IoT security: protocol-level attack-surface modeling (OCPP, MQTT, CAN), ML-based DoS detection, OT/IT convergence.
- Machine learning for intrusion detection: hybrid signature-plus-anomaly fusion, supervised (RandomForest) and unsupervised (IsolationForest) ensembles, false-positive reduction at production scale.
- Medical-device and health-IT security: HL7/FHIR pipeline integrity, IoT patient monitoring, PHI-aware data engineering.
- Detection engineering at scale: soak-tested IDS pipelines, telemetry correlation, AI-assisted alert triage and explanation.
- Supply-chain, compliance, and risk automation: NIST CSF mapping, secure-SaaS backend engineering, governance-by-default platforms.

## EDUCATION

**Sacred Heart University, Fairfield, CT** — Mar 2025 · GPA 3.667/4.0

*M.S. in Cyber/Computer Forensics & Counterterrorism*

- Coursework: digital forensics, counterterrorism, cyber risk, network security.
- Capstone: first-author IEEE GCAIoT 2025 publication, "Plugged-in and Protected: Leveraging ML to Secure IoT-Based EV Charging Stations from DoS Threats" (DOI: 10.1109/GCAIoT68269.2025.11275540).

**Koneru Lakshmaiah (KL) College of Engineering, Guntur, India** — May 2021 · CGPA 8.21/10

*B.Tech in Electronics and Communications Engineering*

- Coursework: signal processing, embedded systems, communication networks, control systems.

## PUBLICATIONS

1. A. S. Shaik, M. Gutla, A. P. R. Kagitala, and S. Bhatia, "Plugged-in and Protected: Leveraging Machine Learning to Secure IoT-Based Electric Vehicle Charging Stations from Denial-of-Service Threats," IEEE Global Conference on Artificial Intelligence and Internet of Things (GCAIoT), CT, USA, 2025. First author. DOI: 10.1109/GCAIoT68269.2025.11275540
   - Trained and benchmarked four supervised ML classifiers (Decision Tree, Random Forest, SVM-RBF, XGBoost) on a 1.62M-sample CICIDS-derived EVCS traffic dataset (benign plus DoS, SMOTE-balanced, 80/20 split) to detect DoS attacks on OCPP-protocol IoT EV charging infrastructure.
   - Random Forest best: 99.03% accuracy, 91.21% precision, 86.56% recall, 88.83% F1, 99.81% AUROC; framed within critical-infrastructure protection (smart grid and clean energy).
2. N. P. Singh, A. Kanakamalla, A. S. Shaik, G. D. Sai, and S. Suman, "Remote Monitoring System of Heart Conditions for Elderly Persons with ECG Machine Using IoT Platform," Journal of Information Systems and Telecommunication (JIST), vol. 10, no. 37, pp. 11-19, Winter 2022. Co-author. DOI: 10.52547/jist.15692.10.37.11
   - Designed and validated an end-to-end IoT pipeline (AD8232 ECG sensor to Arduino/ESP8266 to Wi-Fi to web dashboard) for real-time, unlimited-distance remote cardiac monitoring; auto-flags anomalous ECG and triggers clinician booking without human intervention.

## RESEARCH & PROJECT EXPERIENCE

**Vector, Local-first Windows Blue-Team Security Platform**

*Independent research project · Python, FastAPI, Ollama, Gemini, OpenAI, Grok, SQLite, scapy*

- Designed and shipped a FastAPI control plane (~84 production files) fusing Windows host telemetry with Universal NIDS network alerts; runs SIEM-style correlation producing scored incidents (low/medium/high) with confidence ratings, evidence highlights, timelines, and suggested next steps.
- Built a multi-provider AI router (AIProviderRouter) across Ollama (local default), Gemini, OpenAI, and Grok with automatic fallback, agent-tool bridging, health-check probing, and per-task provider/model attribution.
- Engineered FastAPI security middleware (LocalOnly, API key validation, 120 req/min rate limiting) across 7 API route groups; cut telemetry latency from 3,411 to 2,500 ms by replacing Get-NetAdapter with a Win32_NetworkAdapter CIM query.
- Built a threat-intel ingestion pipeline consuming the curated stamparm/ipsum feed (frequency 3+ filter, 6-hour TTL cache, idempotent persistence).

**Universal NIDS, Hybrid Intrusion-Detection Platform**

*Independent research project · Python, scikit-learn, scapy, FastAPI, Postgres/JSONL, pytest*

- Built a 4-modality detection fusion engine (signature 0.4 + statistical 0.2 + supervised 0.3 + unsupervised 0.1) using IsolationForest and RandomForest; behavior-based detection for campaign, exfiltration, and lateral-movement patterns.
- Reached research Release Candidate status: 195 pytest items, 187 passing, 79.16% coverage (above a 72% enforced floor).
- Production-validated via 6-hour soak test: 87,533 flows, 0 false alerts at peak 107.0% CPU / 409 MB RSS / 13.3 s restart latency.
- Designed a bounded adversary-emulation framework (7 safe scenarios: port scan, brute force, beaconing, exfiltration, lateral sequence, protocol anomaly, campaign chain) with guardrails restricting execution to offline replay or explicitly configured lab CIDRs.

**Career Agent, Bounty Automator, Blockchain Auditor, Asset Monitor**
*Independent research projects · Python, Nuclei, Gemini Flash, headless Edge, Tor, blockchain APIs*

- Career Agent: end-to-end automation scoring JDs 1-10 against a candidate profile, tailoring resumes per JD, rendering PDFs via headless Edge, dispatching via SMTP, and logging via CareerTracker.
- Bounty Automator: AI-augmented pipeline orchestrating Nuclei scans against authorized targets, enriching findings with Gemini-Flash (CVSS 3.1 severity, exploitability, remediation), audit trail to JSONL.
- Blockchain Auditor and Dark-Web Intelligence: investigates BTC/ETH addresses surfaced by a Tor-proxy scout cycle; flags whale wallets (>100 BTC) and correlates findings into a security-event database.

## PROFESSIONAL EXPERIENCE

**Solstice Solutions, Inc.** New Haven, CT · Oct 2025 - Present
*Jr. Data Analyst (Healthcare PHI Integration)*

- Build and maintain HL7 v2 to FHIR R4 integration pipelines on Azure across a 3-stage flow (ingest, transform, export) covering Medicaid and Medicare Part A/B/C; pipelines hold a 99.9% claims approval rate.
- Process and validate EDI healthcare transactions (837 claims, 835 remittance) via SQL-driven ETL with structural and FHIR-conformance validation gates at each stage.
- Develop Power BI dashboards surfacing pipeline health, transaction volumes, and data-quality metrics for clinical and operational stakeholders.

**Infosysnow Inc.** Remote · Aug 2025 - Oct 2025
*Cyber Security Engineer*

- Operated across 3 concurrent security control streams: network hardening, anomaly triage, and remediation tracking; applied OSI-layer analysis to monitor traffic and drive corrective actions.
- Conducted vulnerability assessments and supported penetration testing; created and updated security protocols to reduce organizational risk.

**iQ4** Remote · May 2024 - Aug 2024
*IT Risk Analyst Intern*

- Conducted a 350-hour team NIST CSF analysis of the Uber 2016 exfiltration (AWS S3 + GitHub credential compromise, 57M users); mapped control failures across all 5 CSF functions and produced a 15-slide remediation briefing.
- Identified critical gaps in MFA enforcement, RBAC, credential management, and real-time monitoring; tagged findings to CIA risk categories.

**Sprinto Technology Pvt Ltd** India · Apr 2023 - Dec 2023
*Associate Software Engineer*

- Backend feature development and maintenance on a compliance-focused SaaS (SOC 2 / ISO 27001 family); debugging, testing, and validating components for stable, secure releases.

**Vodafone Idea Ltd.** Hyderabad, India · Nov 2021 - Apr 2023
*Graduate Engineering Trainee to Network Engineer (Operations)*

- Led and supported OSS-to-SRAN migration for 3,500+ sites across 2G/3G/4G, ensuring configuration integrity and minimal service disruption; contributed to a nationwide 4G upgrade across 15 states with second-level BSC fault troubleshooting.

**SanRidge Pvt. Ltd.** Hyderabad, India · Dec 2020 - May 2021
*Android Development Intern*

- Developed Android applications (Kotlin, Android Studio) with Material Design UI through the full SDLC: design, development, testing, debugging.

## TECHNICAL SKILLS

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|---|---|
| **Security** | NIST CSF (deep), MITRE ATT&CK, threat modeling, vulnerability assessment, penetration testing, intrusion detection (4 modalities), anomaly detection / ML (IsolationForest, RandomForest), attack-surface management, dark-web OSINT, blockchain forensics, SIEM, incident response |
| **Data & Health** | HL7 v2, FHIR R4, ETL pipeline design, SQL, Python (pandas, scikit-learn, scapy), Tableau, Power BI, healthcare interoperability, data validation |
| **Engineering** | Python (deep), FastAPI (deep), PowerShell, Bash, Linux, Git, Docker, React, REST APIs, Ollama, AWS, Azure, TCP/IP, DNS, DHCP |
| **Telecom / RAN** | RAN 2G/3G/4G (deep, 3,500+ site migration, 15 states), OSS scripting, XML automation |
| **Tools** | Wireshark, Nmap, Nuclei, subfinder, httpx, scapy, Gitleaks, Gemini / OpenAI / Grok APIs, Tor proxy, Blockchain.info / Etherscan APIs |

## CERTIFICATIONS

- CompTIA Security+ (SY0-701).
- Fortinet NSE 1, NSE 2, and NSE 4 (FortiGate Security 7.0).
- Automation Anywhere Essentials RPA Professional.
- EV Design & Development (ISIEINDIA / MoNRE, Government of India).
- HackerRank: Python (Basic), Problem Solving (Basic and Intermediate).

## HONORS & SERVICE

- Management Conference 2022, Vodafone Idea Ltd. (Apr 2022 - Apr 2023): organizing committee member supporting operational logistics and stakeholder coordination for company-wide leadership development events.